Proof of Theorem 2

Our analysis relies on a sequence of theoretical results analogous to Lemmas $1-7$ and Theorems 1 and 3 in Shah, Xie, and Xu (2022). As the algebraic structure remains largely invariant, we omit repetitive derivations. The key deviations arise from our definition of an optimal conditional VOI action and the selection strategy, both of which depend on κ, propagating changes to the constants in the final results. We provide the theoretical statements with updated variables below, but detail the full proofs only where the deviation is non-trivial.

We first start with the analysis of a class of nonstationary multi-armed bandit as defined in Shah, Xie, and Xu (2022). Let there be $K \geq 1$ arms or actions. Let $\bar{X}_{i,n} = \frac{1}{n}\sum_{t=1}^{n} X_{i,t}$ denote the empirical average of choosing arm i for n times, and let $\mu_{i,n}$ be its expectation. For each arm $i \in [K]$, the reward $X_{i,t}$ is bounded in $[-R, R]$ for some $R > 0$, with the reward sequence being a nonstationary process satisfying:

A. Convergence: The expectation $\mu_{i,n}$ converges to a value μ_i, that is,

$$\mu_i = \lim_{n\to\infty} \mathbb{E}[\bar{X}_{i,n}] \tag{35}$$

B. Concentration: There exist constants $\beta > 1$, $\xi > 0$, and $1/2 \leq \eta < 1$ such that for any $n \geq 1$ and $z \geq 1$:

$$\mathbb{P}\big(n(\bar{X}_{i,n} - \mu_i) \geq n^{\eta} z\big) \leq \beta z^{-\xi}, \tag{36}$$

$$\mathbb{P}\big(n(\bar{X}_{i,n} - \mu_i) \leq -n^{\eta} z\big) \leq \beta z^{-\xi}. \tag{37}$$

Distinct from Shah, Xie, and Xu (2022), we keep track of the modified arm mean biased by a negative fraction κ of its absolute value:

$$\mu_i^U \equiv \mu_i - \kappa_i|\mu_i|, \tag{38}$$

We define the new conditional VOI optimal value with respect to the κ-biased converged expectation

$$\mu_*^U \equiv \max_{i\in[K]} \mu_i^U, \tag{39}$$

where $\kappa_i \in [0, 1]$ may be different for each arm. This will be useful for our VOI-POMDP since it allows open-loop actions to be optimal if their expected reward is within a κ-fraction of their closed-loop counterparts. We define $i^* \in \arg\max_{i\in[K]} \mu_i^U$ and assume the optimal arm is unique. Let the conditional VOI-based suboptimality gap for a suboptimal arm i be:

$$\Delta_i^U \equiv \mu_*^U - \mu_i^U > 0. \tag{40}$$

and the minimum gap $\Delta_{min}^U = min_{i\in[K], i\neq i^*} \Delta_i^U$.

Consider the VOI-UCB:

$$U_i(s_i, t) \equiv \bar{X}_{i,s_i} - \kappa_i|\bar{X}_{i,s_i}| + B_{t,s_i}. \tag{41}$$

The bonus or *bias* B_{t,s_i} is specified as:

$$B_{t,s_i} = \frac{\beta^{1/\xi} \cdot t^{\alpha/\xi}}{s_i^{1-\eta}}, \tag{42}$$

where t is the total number of pulls and s_i is the number of pulls of arm i. Then we define the VOI-UCB algorithm as:

$$I_t \in \arg\max_{i\in[K]} U_i(s_i, t), \tag{43}$$

where a tie between arms is broken arbitrarily. Let $\delta_{i^*,n} = \mu_{i^*,n} - \mu_{i^*}$, which measures how fast the mean of the VOI-optimal arm converges. For simplicity, we sometimes drop the arm index i from quantities related to the optimal arm i^*, e.g. $\delta_{i^*,n} = \delta_{*,n}$.

With the following useful Lemma, we have that the probability that a suboptimal arm or action has a large upper confidence is polynomially small.

Lemma 1. *Let $i \in [K]$ be a suboptimal arm. Define*

$$\begin{aligned} A_i(t) &\equiv \min_{u\in\mathbb{N}} \Big\{ B_{t,u} \leq \frac{\Delta_i^U}{2(1+\kappa_i)} \Big\} \\ &= \left\lceil \left(\frac{2(1+\kappa_i)}{\Delta_i^U} \cdot \beta^{1/\xi} \cdot t^{\alpha/\xi} \right)^{\frac{1}{1-\eta}} \right\rceil \end{aligned} \tag{44}$$

For any $t \geq 1$, $t \geq s_i \geq A_i(t)$, we have

$$\mathbb{P}\big(U_i(s_i, t) > \mu_*^U\big) \leq t^{-\alpha}. \tag{45}$$

Proof. Fix $t \geq 1$ and $s_i \geq A_i(t)$. Define $f(x) = x - \kappa_i|x|$ and note that $\mu_i^U = f(\mu_i)$ and the exploitation component of U_i satisfies $\bar{X}_{i,s_i} - \kappa_i|\bar{X}_{i,s_i}| = f(\bar{X}_{i,s_i})$. We can decompose

$$U_i(s_i, t) - \mu_*^U = \big[f(\bar{X}_{i,s_i}) - f(\mu_i)\big] + B_{t,s_i} - \Delta_i^U.$$

Since $s_i \geq A_i(t)$, by definition $B_{t,s_i} \leq \Delta_i^U/2(1+\kappa_i) \leq \Delta_i^U/2$, so

$$B_{t,s_i} - \Delta_i^U \leq -\frac{\Delta_i^U}{2}.$$

Therefore, $U_i(s_i, t) > \mu_*^U$ requires $f(\bar{X}_{i,s_i}) - f(\mu_i) > \Delta_i^U/2 > 0$. Since f is non-decreasing, the strict inequality $f(\bar{X}_{i,s_i}) - f(\mu_i) > \Delta_i^U/2 > 0$ implies $\bar{X}_{i,s_i} > \mu_i$. Furthermore, for any $y > x$, we have the Lipschitz bound

$$f(y) - f(x) \leq (1+\kappa_i)(y - x),$$

since the steepest slope of f is $1 + \kappa_i$. Applying this with $y = \bar{X}_{i,s_i}$ and $x = \mu_i$ yields

$$f(\bar{X}_{i,s_i}) - f(\mu_i) \leq (1+\kappa_i)(\bar{X}_{i,s_i} - \mu_i).$$

Thus, whenever $U_i(s_i, t) > \mu_*^U$ we must have

$$(1+\kappa_i)(\bar{X}_{i,s_i} - \mu_i) > \frac{\Delta_i^U}{2},$$

i.e.,

$$\bar{X}_{i,s_i} - \mu_i > \frac{\Delta_i^U}{2(1+\kappa_i)} \geq B_{t,s_i},$$

where the last inequality follows from $s_i \geq A_i(t)$. Therefore,

$$\mathbb{P}\big(U_i(s_i, t) > \mu_*^U\big) \leq \mathbb{P}\Big(\bar{X}_{i,s_i} - \mu_i > B_{t,s_i}\Big) \leq t^{-\alpha},$$

using the concentration assumption (37). □

Lemma 2. *For any $t \geq 1$ and $1 \leq s_* \leq t$,*

$$\mathbb{P}\big(U_*(s_*,t) \leq \mu_*^U\big) \leq (1+\kappa_*)^\xi t^{-\alpha},$$

provided β is chosen large enough so that $\beta^{1/\xi} \geq 1+\kappa_*$.

Proof. Define

$$f_*(x) \equiv x - \kappa_*|x|.$$

Then f_* is nondecreasing on $\mathbb{R}$ and

$$U_*(s_*,t) = f_*(\bar{X}_{*,s_*}) + B_{t,s_*}, \qquad \mu_*^U = f_*(\mu_*).$$

Hence

$$\mathbb{P}\big(U_*(s_*,t) \leq \mu_*^U\big) = \mathbb{P}\big(f_*(\bar{X}_{*,s_*}) + B_{t,s_*} \leq f_*(\mu_*)\big).$$

On this event, since $B_{t,s_*} > 0$, we must have $f_*(\bar{X}_{*,s_*}) < f_*(\mu_*)$, and therefore, by monotonicity of f_*,

$$\bar{X}_{*,s_*} \leq \mu_*.$$

Now, for any $x \leq y$,

$$f_*(y) - f_*(x) \leq (1+\kappa_*)(y-x),$$

since the maximum slope of f_* is $1+\kappa_*$. Applying this with $x = \bar{X}_{*,s_*}$ and $y = \mu_*$ gives

$$f_*(\mu_*) - f_*(\bar{X}_{*,s_*}) \leq (1+\kappa_*)(\mu_* - \bar{X}_{*,s_*}).$$

Therefore, whenever $f_*(\bar{X}_{*,s_*}) + B_{t,s_*} \leq f_*(\mu_*)$, we have

$$B_{t,s_*} \leq f_*(\mu_*) - f_*(\bar{X}_{*,s_*}) \leq (1+\kappa_*)(\mu_* - \bar{X}_{*,s_*}),$$

which implies

$$\bar{X}_{*,s_*} - \mu_* \leq -\frac{B_{t,s_*}}{1+\kappa_*}.$$

Thus,

$$\mathbb{P}\big(U_*(s_*,t) \leq \mu_*^U\big) \leq \mathbb{P}\left(\bar{X}_{*,s_*} - \mu_* \leq -\frac{B_{t,s_*}}{1+\kappa_*}\right).$$

Multiplying by s_* and matching the concentration form in (37), set

$$z = s_*^{1-\eta}\frac{B_{t,s_*}}{1+\kappa_*} = s_*^{1-\eta}\frac{\beta^{1/\xi}t^{\alpha/\xi}}{s_*^{1-\eta}(1+\kappa_*)} = \frac{\beta^{1/\xi}}{1+\kappa_*}t^{\alpha/\xi}.$$

If $\beta^{1/\xi} \geq 1+\kappa_*$, then $z \geq 1$ for all $t \geq 1$. Applying the lower-tail concentration assumption (37),

$$\begin{aligned}\mathbb{P}\big(U_*(s_*,t) \leq \mu_*^U\big) \leq \beta z^{-\xi} =& \beta\left(\frac{\beta^{1/\xi}}{1+\kappa_*}t^{\alpha/\xi}\right)^{-\xi} \\ =&(1+\kappa_*)^\xi t^{-\alpha}.\end{aligned}$$

□

Lemma 1 and 2 allow us to upper bound the expected number of suboptimal arm pulls as follows.

Lemma 3 (Adapted from Lemma 2 of (Shah, Xie, and Xu 2022))**.** *Let $i \in [K]$, $i \neq i^*$ be a suboptimal arm. Then, for all $n \geq 1$, the expected number of pulls of arm i satisfies*

$$\mathbb{E}[T_i(n)] \leq \Big(\frac{2(1+\kappa_i)}{\Delta_i^U}\beta^{1/\xi}\Big)^{1/(1-\eta)} n^{\alpha/(\xi(1-\eta))} + \frac{1+(1+\kappa_{i^*})^\xi}{\alpha-2} + 1.$$

Proof. The proof follows Shah, Xie, and Xu (2022)(cf. Lemma 2) after replacing Shah's threshold $A_i(t)$ by our threshold in (44), replacing Shah's suboptimal-arm upper-tail bound by Lemma 1, and replacing Shah's optimal-arm lower-tail bound by Lemma 2. With these substitutions, the decomposition at $T_i(t) \geq A_i(n)$, the summation over polynomial tails, and the conversion to the expected-pulls bound proceed identically up to constants. □

Let $\bar{X}_n \equiv \frac{1}{n}\sum_{i=1}^K T_i(n)\bar{X}_{i,T_i(n)}$ denote the empirical average under the VOI-UCB algorithm (43). Then, $\bar{X}_n$ satisfies the following convergence and concentration properties.

Theorem 4 (Adapted from Theorem 3 of (Shah, Xie, and Xu 2022))**.** *Consider a nonstationary MAB satisfying* (35) *and* (37)*. Suppose that Algorithm* (43) *is applied with parameter α such that $\xi\eta(1-\eta) \leq \alpha < \xi(1-\eta)$ and $\alpha > 2$. Then, the following holds:*

***A. Convergence**:*

$$\big|\mathbb{E}[\bar{X}_n] - \mu_*\big| \leq |\delta_{*,n}| + \frac{2R(K-1)}{n}\Big(\big(\frac{4}{\Delta_{min}^U}\beta^{\frac{1}{\xi}}\big)^{\frac{1}{1-\eta}} \cdot n^{\frac{\alpha}{(\xi(1-\eta))}} + \frac{1+2^\xi}{\alpha-2} + 1\Big),$$

***B. Concentration**: there exists constants, $\beta' > 1$ and $\xi' > 0$ and $1/2 \leq \eta' < 1$ such that for every $n \geq 1$ and every $z \geq 1$,*

$$\mathbb{P}(n\bar{X}_n - n\mu_* \geq n^{\eta'}z) \leq \frac{\beta'}{z^{\xi'}},$$

$$\mathbb{P}(n\bar{X}_n - n\mu_* \leq -n^{\eta'}z) \leq \frac{\beta'}{z^{\xi'}},$$

where $\eta' = \frac{\alpha}{\xi(1-\eta)}, \xi' = \alpha - 1, \beta'$ depends on $R, K, \Delta_{min}^U, \beta, \xi, \alpha, \eta, \kappa$.

Proof. The proof follows the proof of Theorem 3 in Shah, Xie, and Xu (2022) after replacing Shah's threshold by

$$A(t) = \max_{i\in[K]} A_i(t) = \left\lceil \left(\frac{4}{\Delta_{\min}^U}\beta^{1/\xi}\right)^{\frac{1}{1-\eta}} t^{\frac{\alpha}{\xi(1-\eta)}} \right\rceil,$$

replacing Shah's suboptimal-arm upper-tail bound by Lemma 1, replacing Shah's optimal-arm lower-tail bound by Lemma 2, and using Lemma 3 in place of Shah's expected-pulls bound. With these substitutions, the remainder of the convergence and concentration argument is identical up to constants, yielding the stated result. □

We aim to prove that under our VOI-based MCTS (adapted from Algorithm 1 in Shah, Xie, and Xu (2022)), the mean of the empirical reward at the root node of the MCTS tree is within $O(n^{\eta-1})$ of the mean reward obtained as the number of samples tends to infinity. To construct the VOI-based MCTS, we replace the standard action selection policy (cf. (5) of Shah, Xie, and Xu (2022)) with the following κ-biased policy:

$$a^{(h+1)} = \arg\max_{a\in\mathcal{A}} Q_N^h(s,a) - \kappa_a|Q_N^h(s,a)| + B_N^h(s,a) \tag{46}$$

where

$$Q_N^h(s,a) = \frac{q^{(h+1)}(s^{(h)},a) + \gamma \tilde{v}^{(h+1)}(s^{(h)} \circ a)}{N^{(h+1)}(s^{(h)} \circ a)} \quad (47)$$

and

$$B_N^h(s,a) = \frac{(\beta^{(h+1)})^{1/\xi^{(h+1)}} \cdot (N^{(h)}(s^{(h)}))^{\alpha^{(h+1)}/\xi^{(h+1)}}}{(N^{(h+1)}(s^{(h)} \circ a))^{1-\eta^{(h+1)}}} \quad (48)$$

Note, κ_a is parameterized by action a where for each $a \in [K]$, $\kappa_a \in [0,1]$

Analyzing Leaf Level $d = D$ Let n_{D-1} denote the number of nodes at depth $D-1$, corresponding to states $s_{1,D-1}, \ldots, s_{n_{D-1},D-1} \in \mathcal{S}$. Consider node $i \in [n_{D-1}]$ at level $D-1$ corresponding to state $s_{i,D-1}$. Under the VOI-MCTS algorithm, whenever this node is visited, an action $a \in [K]$ will be taken, reaching the leaf node $s'_D = s_{i,D-1} \circ a$. This results in the cumulative reward $\mathcal{R}(s_{i,D-1}, a) + \gamma \tilde{v}^{(D)}(s'_D)$, where $\tilde{v}^{(D)}(\cdot)$ denotes an initial value estimate $V_{\epsilon_0} = 0$. Similar to (38) and (39) for the bandit, we define the following VOI-based terms for the tree:

$$\begin{aligned}
\mu_a^{(D\text{-}1)}(s_{i,D\text{-}1}) =& \mathbb{E}\left[\mathcal{R}(s_{i,D\text{-}1}, a)\right] + \gamma \tilde{v}^{(D)}(s_{i,D\text{-}1} \circ a), \\
\mu_a^{U,(D\text{-}1)}(s_{i,D\text{-}1}) =& \mu_a^{(D\text{-}1)}(s_{i,D\text{-}1}) - \kappa_a^{(D\text{-}1)} |\mu_a^{(D\text{-}1)}(s_{i,D\text{-}1})|, \\
a_*^{(D\text{-}1)}(s_{i,D\text{-}1}) \in& \arg\max_{a \in [K]} \mu_a^{U,(D\text{-}1)}(s_{i,D\text{-}1}), \\
\mu_*^{U,(D\text{-}1)}(s_{i,D\text{-}1}) =& \max_{a \in [K]} \mu_a^{U,(D\text{-}1)}(s_{i,D\text{-}1}), \\
\mu_*^{(D\text{-}1)}(s_{i,D\text{-}1}) =& \mu^{(D\text{-}1)}_{a_*^{(D\text{-}1)}(s_{i,D\text{-}1})}(s_{i,D\text{-}1}), \\
\Delta_{min}^{U,(D\text{-}1)}(s_{i,D\text{-}1}) =& \mu_*^{U,(D\text{-}1)}(s_{i,D\text{-}1}) \\
& - \max_{a \neq a_*^{(D\text{-}1)}(s_{i,D\text{-}1})} \mu_a^{U,(D\text{-}1)}(s_{i,D\text{-}1}),
\end{aligned} \quad (49)$$

where all rewards belong to $[-\tilde{R}_{\max}^{(D-1)}, \tilde{R}_{\max}^{(D-1)}]$ consistent with Shah, Xie, and Xu (2022).

By applying Lemma 4 from Shah, Xie, and Xu (2022), we obtain that for $\eta^{(D)} \in \left[\frac{1}{2}, 1\right)$, and sufficiently large $\xi^{(D)}$, there exists a constant $\beta^{(D)}$ such that the collected rewards at $s_{i,D-1}$ (i.e., the sum of the i.i.d. stochastic reward and the value evaluation) satisfy the convergence property (cf. (35)) and the concentration property (cf. (37)). Therefore, by an application of Theorem 4, we have the following Lemma.

Lemma 4 (Adapted from Lemma 5 of Shah, Xie, and Xu (2022))**.** *Consider a node corresponding to state $s_{i,D-1}$ at level $d = D-1$ within the VOI-MCTS for $i \in [n_{D-1}]$. Let $\tilde{v}^{(D-1)}(s_{i,D-1})_n$ be the total discounted reward collected at $s_{i,D-1}$ during $n \geq 1$ visits of it, to one of its K leaf nodes under the VOI-UCB policy. Then, for the choice of appropriately large $\beta^{(D)} > 0$, for a given $\xi^{(D)} > 0$, $\eta^{(D)} \in [\frac{1}{2}, 1)$ and $\alpha^{(D)} > 2$, we have*

A. Convergence:

$$\left| \mathbb{E}\left[\frac{1}{n} \tilde{v}^{(D-1)}(s_{i,D-1})_n\right] - \mu_*^{(D-1)}(s_{i,D-1}) \right|$$
$$\leq \psi_n^{(D-1)} \left(\left(\frac{4(\beta^{(D)})^{\frac{1}{\xi^{(D)}}}}{\Delta_{min}^{U,(D-1)}(s_{i,D-1})} \right)^{\frac{1}{1-\eta^{(D)}}} \cdot n^{\frac{\alpha^{(D)}}{\xi^{(D)}(1-\eta^{(D)})}} + C^{(D)} \right),$$

where $\psi_n^{(D-1)} = \frac{2\tilde{R}_{\max}^{(D-1)}(K-1)}{n}$ and $C^{(D)} = \frac{1+2^{\xi^{(D)}}}{\alpha^{(D)}-2} + 1$.

B. Concentration: *there exist constants, $\beta' > 1$ and $\xi' > 1$ and $1/2 \leq \eta' < 1$ such that for every $n \geq 1$ and every $z \geq 1$,*

$$\mathbb{P}(\tilde{v}^{(D-1)}(s_{i,D-1})_n - n\mu_*^{(D-1)}(s_{i,D-1}) \geq n^{\eta'} z) \leq \frac{\beta'}{z^{\xi'}},$$
$$\mathbb{P}(\tilde{v}^{(D-1)}(s_{i,D-1})_n - n\mu_*^{(D-1)}(s_{i,D-1}) \leq -n^{\eta'} z) \leq \frac{\beta'}{z^{\xi'}},$$

where $\eta' = \frac{\alpha^{(D)}}{\xi^{(D)}(1-\eta^{(D)})}$, $\xi' = \alpha^{(D)} - 1$, and β' is a large enough constant that is function of parameters $\alpha^{(D)}, \beta^{(D)}, \xi^{(D)}, \eta^{(D)}, \tilde{R}_{\max}^{(D-1)}, K$, and $\Delta_{\min}^{U,(D-1)}(s_{i,D\text{-}1})$.

Recursion: Going from level d to $d-1$ The proof for the convergence and concentration properties of the value estimates across all levels d of the VOI-MCTS tree proceeds via mathematical induction, following the structure established by Shah, Xie, and Xu (2022). The mechanism of the induction is identical to the original paper, but the final, global convergence rate differs due to how we defined the sample threshold $A_i(t)$.

Assume the estimated value $\tilde{v}^{(d)}(\cdot)$ at level d satisfies properties (35) and (37) with parameters $\alpha^{(d)}, \xi^{(d)}, \eta^{(d)}$, and an appropriately large constant $\beta^{(d)}$. For any node $s_{i,d-1}$ at level $d-1$, the effective reward received is the combined value $\mathcal{R}(s_{i,d-1}, a) + \gamma \tilde{v}^{(d)}(s'_d)$ where $s'_d = s_{i,d-1} \circ a$, and $\tilde{v}^{(d)}(\cdot)$ returns the collected rewards recursively until the leaf node at depth D. By the inductive hypothesis and an application of Lemma 4 (Shah, Xie, and Xu 2022), this combined reward sequence continues to satisfy assumptions (35) and (37) with the same α, ξ, η dependencies and a re-adjusted constant $\beta^{(d-1)}$.

Note, $\mu_*^{(d-1)}(s_{i,d-1}), \mu_a^{U,(d-1)}(s_{i,d-1}), \mu_*^{U,(d-1)}(s_{i,d-1})$, $a_*^{(d-1)}(s_{i,d-1})$, and $\Delta_{\min}^{U,(d-1)}(s_{i,d-1})$ are defined similarly to (49), and rewards here belong to $[-\tilde{R}_{\max}^{(d-1)}, \tilde{R}_{\max}^{(d-1)}]$. By an application of Theorem 4, we obtain the following result regarding the desired convergence and concentration properties of the combined reward sequence at level $d-1$, which completes the induction and proves the results for all $d \geq 0$.

Lemma 5 (Adapted from Lemma 6 of Shah, Xie, and Xu (2022))**.** *Consider a node corresponding to state $s_{i,d-1}$ at level $d-1$ within the VOIMCP for $i \in [n_{d-1}]$. Let $\tilde{v}^{(d-1)}(s_{i,d-1})_n$ be the total discounted reward collected at $s_{i,d-1}$ during $n \geq 1$ visits. Then, for the choice of appropriately large $\beta^{(d)} > 0$, for a given $\xi^{(d)} > 0$, $\eta^{(d)} \in [\frac{1}{2}, 1)$ and $\alpha^{(d)} > 2$, we have*

A. Convergence:

$$\left|\mathbb{E}\left[\frac{1}{n}\tilde{v}^{(d-1)}(s_{i,d-1})_n\right] - \mu_*^{(d-1)}(s_{i,d-1})\right|$$
$$\leq \psi_n^{(d-1)}\left(\left(\frac{4(\beta^{(d)})^{\frac{1}{\xi^{(d)}}}}{\Delta_{\min}^{U,(d-1)}(s_{i,d-1})}\right)^{\frac{1}{1-\eta^{(d)}}} \cdot n^{\frac{\alpha^{(d)}}{\xi^{(d)}(1-\eta^{(d)})}} + C^{(d)}\right),$$

where $\psi_n^{(d-1)} = \frac{2\tilde{R}_{\max}^{(d-1)}(K-1)}{n}$ *and* $C^{(d)} = \frac{1+2^{\xi^{(d)}}}{\alpha^{(d)}-2} + 1$.

B. Concentration: *there exist constants,* $\beta' > 1$ *and* $\xi' > 1$ *and* $1/2 \leq \eta' < 1$ *such that for* $n \geq 1$, $z \geq 1$,

$$\mathbb{P}(\tilde{v}^{(d-1)}(s_{i,d-1})_n - n\mu_*^{(d-1)}(s_{i,d-1}) \geq n^{\eta'}z) \leq \frac{\beta'}{z^{\xi'}},$$
$$\mathbb{P}(\tilde{v}^{(d-1)}(s_{i,d-1})_n - n\mu_*^{(d-1)}(s_{i,d-1}) \leq -n^{\eta'}z) \leq \frac{\beta'}{z^{\xi'}},$$

where $\eta' = \frac{\alpha^{(d)}}{\xi^{(d)}(1-\eta^{(d)})}$, $\xi' = \alpha^{(d)} - 1$, *and* β' *is a large enough constant that is function of parameters* $\alpha^{(d)}, \beta^{(d)}, \xi^{(d)}, \eta^{(d)}, \tilde{R}_{\max}^{(d-1)}, K, \Delta_{\min}^{U,(d-1)}(s_{i,d-1})$.

Completing Proof of Theorem 2 Using Lemma 5, we conclude that the recursive relationship going from d to $d-1$ holds for all $d \geq 1$, with level 0 being the root. Thus, for the root state $s^{(0)}$, under n total simulations of VOI-MCTS, the empirical average of the rewards $\frac{1}{n}\tilde{v}^{(0)}(s_0)_n$ satisfies

$$\left|\mathbb{E}\left[\frac{1}{n}\tilde{v}^{(0)}(s_0)_n\right] - \mu_*^{(0)}(s_0)\right| \leq O\left(n^{\frac{\alpha^{(0)}}{\xi^{(0)}(1-\eta^{(0)})}-1}\right) = O(n^{\eta-1}), \tag{50}$$

where $\mu_*^{(0)}(s_0)$ is the raw depth-D value obtained at the root by the κ-adaptive VOI-MCTS recursion.

To transfer this root-level VOI-MCTS result to the VOIMCP estimator $\bar{V}_n(b_0) = \frac{1}{n}\sum_{a\in\mathcal{A}'} N(h_0a)\bar{Q}(h_0a)$, we introduce the derived history-MDP for the transformed VOI-POMDP $\mathcal{P}'$. Let

$$\tilde{\mathcal{M}}' = (\mathcal{H}', \mathcal{A}', \bar{\mathcal{T}}, \bar{\mathcal{R}}, D, h_0, \gamma),$$

where $\mathcal{H}'$ is the set of reachable histories of $\mathcal{P}'$, and for any $h \in \mathcal{H}'$ and $a \in \mathcal{A}'$,

$$\bar{\mathcal{R}}(h,a) := \sum_{s\in\mathcal{S}} B(s,h)\,\mathcal{R}'(s,a), \tag{51}$$

$$\bar{\mathcal{T}}(h,a,hao) := \sum_{s\in\mathcal{S}}\sum_{s'\in\mathcal{S}} B(s,h)\,\mathcal{T}'(s,a,s')\,\mathcal{Z}'(s',o,a), \tag{52}$$

with $B(\cdot,h)$ denoting the exact belief induced by h in $\mathcal{P}'$. Define $\kappa_i = \kappa$ for closed-loop actions, and $\kappa_i = 0$ for open-loop actions.

The deterministic-transition analysis above extends to stochastic transitions by the same meta-node construction as in Lemma A.1 of Shah, Xie, and Xu (2022). Since $\mathcal{P}'$ is a finite-horizon POMDP and $B(\cdot,h)$ is the exact belief, the derived history-MDP equivalence of Lemmas 1 and 2 in Silver and Veness (2010) applies.

Let $\tilde{V}_D^{\kappa}(h_0)$ denote the depth-D value at the root of $\tilde{\mathcal{M}}'$ under the same κ-adaptive selection rule induced by the VOI-MCTS recursion. Since Lemma 5 is centered at the raw mean of the action selected by this κ-adaptive rule, the root target satisfies

$$\mu_*^{(0)}(s_0) = \tilde{V}_D^{\kappa}(h_0).$$

By Lemma 1 of Silver and Veness (2010), the value of any history-dependent policy in $\mathcal{P}'$ is equal to its value in the derived history-MDP $\tilde{\mathcal{M}}'$. Therefore,

$$\tilde{V}_D^{\kappa}(h_0) = \hat{V}_D'(b_0).$$

Furthermore, by Proposition 1,

$$\hat{V}_D'(b_0) = \hat{V}_D^*(b_0).$$

Moreover, by Lemma 2 of Silver and Veness (2010), the rollout distribution in $\mathcal{P}'$ is equal to the rollout distribution in $\tilde{\mathcal{M}}'$ for any history-dependent simulation policy. Since VOIMCP in $\mathcal{P}'$ and VOI-MCTS in $\tilde{\mathcal{M}}'$ use the same history-dependent selection rule, their simulation traces are equal in distribution. Consequently, the VOIMCP root estimator

$$\bar{V}_n(b_0) = \frac{1}{n}\sum_{a\in\mathcal{A}'} N(h_0a)\bar{Q}(h_0a)$$

has the same distribution as the root empirical average

$$\frac{1}{n}\tilde{v}^{(0)}(s_0)_n$$

analyzed above. Hence,

$$\left|\mathbb{E}[\bar{V}_n(b_0)] - \hat{V}_D^*(b_0)\right| \leq O(n^{\eta-1}),$$

which proves Theorem 2.

Theorem 2 establishes that VOIMCP converges to the adaptive VOI value $\hat{V}_D^*$, while Theorem 1 bounds how far $\hat{V}_D^*$ can deviate from the true optimal value V_D^*. Combining the two yields the following bound.

Corollary 1.

$$\left|\mathbb{E}[\bar{V}_n(b_0)] - V_D^*(b_0)\right| \leq O\big(n^{\eta-1}\big) + \kappa\frac{R_{\max}}{1-\gamma}\left(\frac{1-\gamma^D}{1-\gamma}\right).$$

Proof. By the triangle inequality,

$$\left|\mathbb{E}[\bar{V}_n(b_0)] - V_D^*(b_0)\right| \leq \left|\mathbb{E}[\bar{V}_n(b_0)] - \hat{V}_D^*(b_0)\right| + \left|\hat{V}_D^*(b_0) - V_D^*(b_0)\right|.$$

The first term is $O(n^{\eta-1})$ by Theorem 2, and the second term is bounded by Theorem 1. □

Proof of Theorem 3

The proof of Theorem 3 follows the same structure as Theorem 2. Thus, our analysis relies on a sequence of theoretical results analogous to Lemmas $1-7$ and Theorems 1 and 3 in Shah, Xie, and Xu (2022). From Theorem 2, the main modifications are the optimality criterion and a scheduled decaying κ_i. We provide the statements with updated variables below, but detail the full proofs only where the deviation is non-trivial.

We begin again by considering the same non-stationary multi-armed bandit defined in Shah, Xie, and Xu (2022). Let there be $K \geq 1$ arms or actions. Let $\bar{X}_{i,n} = \frac{1}{n}\sum_{t=1}^{n} X_{i,t}$ denote the empirical average of choosing arm i for n times, and let $\mu_{i,n}$ be its expectation. For each arm $i \in [K]$, the reward $X_{i,t}$ is bounded in $[-R, R]$ for some $R > 0$, with the reward sequence being a nonstationary process satisfying (35) and (37).

Distinct from Theorem 2, we now consider the optimal value with respect to the converged expectation of each arm,

$$\mu_* = \max_{i\in[K]} \mu_i \tag{53}$$

with the suboptimality gap

$$\Delta_i = \mu_* - \mu_i. \tag{54}$$

As stated in the main text, any annealing schedule satisfying $0 \leq \kappa_i(t, s_i) \leq \frac{1}{c_\kappa R} B_{t,s_i}$ suffices. Define the scheduled VOI-UCB dependent on a time-varying κ as

$$U'(s_i, t) \equiv \bar{X}_{i,s_i} - \kappa_i(t, s_i)|\bar{X}_{i,s_i}| + B_{t,s_i}, \tag{55}$$

Then we define the annealed VOI-UCB algorithm as:

$$I_t \in \arg\max_{i\in[K]} U'_i(s_i, t), \tag{56}$$

where a tie between arms is broken arbitrarily. All other terms are as defined in Shah, Xie, and Xu (2022).

With the above, we have that the probability that a suboptimal arm or action has a large upper confidence is polynomially small.

Lemma 6. *Let $i \in [K]$ be a suboptimal arm. Define*

$$A'_i(t) \equiv \min_{u\in\mathbb{N}} \left\{ B_{t,u} \leq \frac{\Delta_i}{2} \right\} = \left\lceil \left(\frac{2}{\Delta_i} \cdot \beta^{1/\xi} \cdot t^{\alpha/\xi} \right)^{\frac{1}{1-\eta}} \right\rceil \tag{57}$$

For any $t \geq 1$, $t \geq s_i \geq A'_i(t)$,

$$\mathbb{P}\big(U'_i(s_i, t) > \mu_*\big) \leq t^{-\alpha}. \tag{58}$$

Proof. Fix $t \geq 1$ and $s_i \geq A'_i(t)$. Since $\kappa_i(t, s_i) \geq 0$, the penalty term $\kappa_i(t, s_i)|\bar{X}_{i,s_i}|$ is non-negative, so

$$U'_i(s_i, t) = \bar{X}_{i,s_i} - \kappa_i(t, s_i)|\bar{X}_{i,s_i}| + B_{t,s_i} \leq \bar{X}_{i,s_i} + B_{t,s_i}.$$

Therefore,

$$\begin{aligned}\mathbb{P}\big(U'_i(s_i, t) > \mu_*\big) &\leq \mathbb{P}\big(\bar{X}_{i,s_i} + B_{t,s_i} > \mu_*\big) \\ &= \mathbb{P}\big(\bar{X}_{i,s_i} - \mu_i > \Delta_i - B_{t,s_i}\big).\end{aligned}$$

Since $s_i \geq A'_i(t)$, by definition $B_{t,s_i} \leq \Delta_i/2$, so $\Delta_i - B_{t,s_i} \geq \Delta_i/2 \geq B_{t,s_i}$. Hence,

$$\mathbb{P}\big(U'_i(s_i, t) > \mu_*\big) \leq \mathbb{P}\big(\bar{X}_{i,s_i} - \mu_i > B_{t,s_i}\big) \leq t^{-\alpha},$$

where the last step uses the concentration assumption (37). □

As in Theorem 4, let $i^* \in \arg\max_{i\in[K]} \mu_i$ be the optimal arm. Then we have the following lemma.

Lemma 7. *For any $t \geq 1$ and $1 \leq s \leq t$,*

$$\mathbb{P}\big(U'_*(s_*, t) \leq \mu_*\big) \leq \Big(\frac{c_\kappa}{c_\kappa - 1}\Big)^{\xi} t^{-\alpha}.$$

Proof. Since $\kappa_*(t, s_*) \leq \frac{1}{c_\kappa R} B_{t,s_*}$ and $|\bar{X}_{*,s_*}| \leq R$, the penalty satisfies

$$\kappa_*(t, s_*)|\bar{X}_{*,s_*}| \leq \frac{1}{c_\kappa R} B_{t,s_*} \cdot R = \frac{B_{t,s_*}}{c_\kappa}.$$

Therefore,

$$B_{t,s_*} - \kappa_*(t, s_*)|\bar{X}_{*,s_*}| \geq B_{t,s_*}\Big(\frac{c_\kappa - 1}{c_\kappa}\Big)$$

Hence,

$$\begin{aligned}&\mathbb{P}\big(U'_*(s_*, t) \leq \mu_*\big) \\ &\quad= \mathbb{P}\big(\bar{X}_{*,s_*} + B_{t,s_*} - \kappa_*(t, s_*)|\bar{X}_{*,s_*}| \leq \mu_*\big) \\ &\quad\leq \mathbb{P}\big(\bar{X}_{*,s_*} + B_{t,s_*}\Big(\frac{c_\kappa - 1}{c_\kappa}\Big) \leq \mu_*\big) \\ &\quad= \mathbb{P}\big(s(\bar{X}_{*,s_*} - \mu_*) \leq -s \cdot B_{t,s_*}\Big(\frac{c_\kappa - 1}{c_\kappa}\Big)\big)\end{aligned}$$

To apply the concentration assumption (37), we match $s^\eta z = s \cdot B_{t,s_*}((c_\kappa - 1)/c_\kappa)$, giving

$$\begin{aligned}z &= s^{1-\eta} B_{t,s_*}\Big(\frac{c_\kappa - 1}{c_\kappa}\Big) = s^{1-\eta} \cdot \frac{\beta^{1/\xi} t^{\alpha/\xi}}{s^{1-\eta}}\Big(\frac{c_\kappa - 1}{c_\kappa}\Big) \\ &= \beta^{1/\xi} t^{\alpha/\xi}\Big(\frac{c_\kappa - 1}{c_\kappa}\Big).\end{aligned}$$

To ensure $z \geq 1$, without loss of generality, let β be large enough that $\beta^{1/\xi} \geq (c_\kappa/(c_\kappa - 1))$ (cf. the discussion below Eq. (20) in Shah, Xie, and Xu (2022)). Applying (37):

$$\begin{aligned}\mathbb{P}\big(U'_*(s_*, t) \leq \mu_*\big) \leq \beta z^{-\xi} &= \beta \cdot \left(\beta^{1/\xi} t^{\alpha/\xi}\Big(\frac{c_\kappa - 1}{c_\kappa}\Big)\right)^{-\xi} \\ &= \Big(\frac{c_\kappa}{c_\kappa - 1}\Big)^{\xi} t^{-\alpha}.\end{aligned}$$

□

Lemma 6 allows us to upper bound the expected number of suboptimal arm pulls as follows.

Lemma 8. *Let $i \in [K]$, $i \neq i^*$ be a suboptimal arm. Then, for all $n \geq 1$, the expected number of pulls of arm i satisfies*

$$\mathbb{E}[T_i(n)] \leq \left(\frac{2}{\Delta_i} \cdot \beta^{1/\xi}\right)^{\frac{1}{1-\eta}} n^{\frac{\alpha}{\xi(1-\eta)}} + \frac{1 + \left(\frac{c_\kappa}{c_\kappa - 1}\right)^{\xi}}{\alpha - 2} + 1.$$

Proof. The proof follows the structure of Lemma 3. Event E_1 (suboptimal arm's VOI-UCB exceeds μ_*) is bounded using Lemma 6. Event E_2 (optimal arm's VOI-UCB falls below μ_*) is bounded using Lemma 7. The remainder follows identically, with the constant $(1 + (c_\kappa/(c_\kappa - 1))^{\xi})/(\alpha - 2)$ absorbing the factor from the scaled bonus. □

Now we can state the convergence and concentration results for the annealed VOI nonstationary multi-armed bandit. Let $\bar{X}_n \equiv \frac{1}{n}\sum_{i=1}^{K} T_i(n)\bar{X}_{i,T_i(n)}$ denote the empirical average under the annealed VOI-UCB algorithm (56). Then, $\bar{X}_n$ satisfies the following convergence and concentration properties.

Theorem 5. *Consider a nonstationary MAB satisfying* (35) *and* (37). *Suppose that Algorithm* (56) *is applied with parameter* α *such that* $\xi\eta(1-\eta) \le \alpha < \xi(1-\eta)$ *and* $\alpha > 2$. *Then, the following holds:*
A. Convergence:

$$\begin{aligned}&\left|\mathbb{E}[\bar{X}_n] - \mu^*\right| \\ &\le |\delta_n^*| \\ &+ \frac{2R(K-1)}{n}\left(\left(\frac{2}{\Delta_{\min}}\beta^{\frac{1}{\xi}}\right)^{\frac{1}{1-\eta}} \cdot n^{\frac{\alpha}{\xi(1-\eta)}} + \frac{1+\left(\frac{c_\kappa}{c_\kappa-1}\right)^{\xi}}{\alpha-2} + 1\right).\end{aligned}$$

B. Concentration: *there exists constants,* $\beta' > 1$ *and* $\xi' > 0$ *and* $1/2 \le \eta' < 1$ *such that for every* $n \ge 1$ *and every* $z \ge 1$,

$$\mathbb{P}(n\bar{X}_n - n\mu_* \ge n^{\eta'} z) \le \frac{\beta'}{z^{\xi'}},$$
$$\mathbb{P}(n\bar{X}_n - n\mu_* \le -n^{\eta'} z) \le \frac{\beta'}{z^{\xi'}},$$

where $\eta' = \frac{\alpha}{\xi(1-\eta)}, \xi' = \alpha - 1, \beta'$ *is a large enough constant that depends on* $R, K, \Delta_{min}, \beta, \xi, \alpha, \eta$.

Proof. The proof follows directly from the proof of Theorem 4, by substituting the global sample threshold:

$$A'(t) = \max_{i\in[K]} A_i'(t) = \left\lceil \left(\frac{2}{\Delta_{\min}} \cdot \beta^{1/\xi}\right)^{1/(1-\eta)} t^{\alpha/(\xi(1-\eta))} \right\rceil. \tag{59}$$

□

Analyzing Leaf Level $d = D$ By applying Lemma 4 from Shah, Xie, and Xu (2022), we obtain that for $\eta^{(D)} \in \left[\frac{1}{2}, 1\right)$, and sufficiently large $\xi^{(D)}$, there exists a constant $\beta^{(D)}$ such that the collected rewards at $s_{i,D-1}$ (i.e., the sum of the i.i.d. stochastic reward and the value evaluation) satisfy the convergence property (cf. (35)) and the concentration property (cf. (37)). The optimal mean target $\mu_*^{(D\text{-}1)}(s_{i,D\text{-}1})$ and $\Delta_{min}^{(D\text{-}1)}(s_{i,D\text{-}1})$ are as defined in Shah, Xie, and Xu (2022). Then, by an application of Theorem 5, we have the following Lemma.

Lemma 9. *Consider a node corresponding to history* $s_{i,D-1}$ *at level* $D-1$ *within the VOIMCP for* $i \in [n_{D-1}]$. *Let* $\tilde{v}^{(D-1)}(s_{i,D-1})_n$ *be the total discounted reward collected at* $s_{i,D-1}$ *during* $n \ge 1$ *visits. Then, for the choice of appropriately large* $\beta^{(D)} > 0$, *for a given* $\xi^{(D)} > 0$, $\eta^{(D)} \in [\frac{1}{2}, 1)$ *and* $\alpha^{(D)} > 2$, *we have*

A. Convergence:

$$\begin{aligned}&\left|E\left[\frac{1}{n}\tilde{v}^{(D-1)}(s_{i,D-1})_n\right] - \mu_*^{(D-1)}(s_{i,D-1})\right| \\ &\le \psi_n^{(D-1)}\left(\left(\frac{2(\beta^{(D)})^{\frac{1}{\xi^{(D)}}}}{\Delta_{\min}^{(D-1)}(s_{i,D-1})}\right)^{\frac{1}{1-\eta^{(D)}}} \cdot n^{\frac{\alpha^{(D)}}{\xi^{(D)}(1-\eta^{(D)})}} + C^{(D)}\right),\end{aligned}$$

where $\psi_n^{(D-1)} = \frac{2\tilde{R}_{\max}^{(D-1)}(K-1)}{n}$ *and* $C^{(D)} = \frac{1+\left(\frac{c_\kappa}{c_\kappa-1}\right)^{\xi^{(D)}}}{\alpha^{(D)}-2} + 1$.

B. Concentration: *there exist constants,* $\beta' > 1$ *and* $\xi' > 1$ *and* $1/2 \le \eta' < 1$ *such that for* $n \ge 1$, $z \ge 1$,

$$\mathbb{P}(\tilde{v}^{(D-1)}(s_{i,D-1})_n - n\mu_*^{(D-1)}(s_{i,D-1}) \ge n^{\eta'} z) \le \frac{\beta'}{z^{\xi'}},$$
$$\mathbb{P}(\tilde{v}^{(D-1)}(s_{i,D-1})_n - n\mu_*^{(D-1)}(s_{i,D-1}) \le -n^{\eta'} z) \le \frac{\beta'}{z^{\xi'}},$$

where $\eta' = \frac{\alpha^{(D)}}{\xi^{(D)}(1-\eta^{(D)})}$, $\xi' = \alpha^{(D)} - 1$, *and* β' *is a large enough constant that is function of parameters* $\alpha^{(D)}, \beta^{(D)}, \xi^{(D)}, \eta^{(D)}, \tilde{R}_{\max}^{(D-1)}, K, \Delta_{\min}^{(D-1)}(s_{i,D-1})$.

Recursion: Going from level d **to** $d-1$ The proof for the convergence and concentration properties of the value estimates across all levels d of the VOIMCP tree proceeds via mathematical induction, following the structure in Shah, Xie, and Xu (2022). The mechanism of the induction is identical to the original paper, but the final, global convergence rate differs due to how we defined the sample threshold $A_i'(t)$.

Assume the estimated value $\tilde{v}^{(d)}(\cdot)$ at level d satisfies properties (35) and (37) with parameters $\alpha^{(d)}, \xi^{(d)}, \eta^{(d)}$, and an appropriately large constant $\beta^{(d)}$. For any node $s_{i,d-1}$ at level $d-1$, the effective reward received is the combined value $R(s_{i,d-1}, a) + \gamma\tilde{v}^{(d)}(s_d')$. By the inductive hypothesis and an application of Lemma 4 of Shah, Xie, and Xu (2022), this combined reward sequence continues to satisfy assumptions (35) and (37) with the same α, ξ, η dependencies and a re-adjusted constant $\beta^{(d-1)}$.

Under the same bounded reward, value definitions, and objects $\mu_*^{(d\text{-}1)}(s_{i,d\text{-}1})$ and $\Delta_{min}^{(d\text{-}1)}(s_{i,d\text{-}1})$ defined in Shah, Xie, and Xu (2022), we apply Theorem 5 and obtain the following result regarding the desired convergence and concentration properties of the combined reward sequence at level $d-1$, which completes the induction and proves the results for all $d \ge 0$.

Lemma 10. *Consider a node corresponding to state* $s_{i,d-1}$ *at level* $d-1$ *within the VOIMCP for* $i \in [n_{d-1}]$. *Let* $\tilde{v}^{(d-1)}(s_{i,d-1})_n$ *be the total discounted reward collected at* $s_{i,d-1}$ *during* $n \ge 1$ *visits. Then, for the choice of appropriately large* $\beta^{(d)} > 0$, *for a given* $\xi^{(d)} > 0$, $\eta^{(d)} \in [\frac{1}{2}, 1)$ *and* $\alpha^{(d)} > 2$, *we have*

A. Convergence:

$$\left|E\left[\frac{1}{n}\tilde{v}^{(d-1)}(s_{i,d-1})_n\right] - \mu_*^{(d-1)}(s_{i,d-1})\right|$$
$$\leq \psi_n^{(d-1)}\left(\left(\frac{2(\beta^{(d)})^{\frac{1}{\xi^{(d)}}}}{\Delta_{\min}^{(d-1)}(s_{i,d-1})}\right)^{\frac{1}{1-\eta^{(d)}}} \cdot n^{\frac{\alpha^{(d)}}{\xi^{(d)}(1-\eta^{(d)})}} + C^{(d)}\right),$$

where $\psi_n^{(d-1)} = \frac{2\tilde{R}_{\max}^{(d-1)}(K-1)}{n}$ *and* $C^{(d)} = \frac{1+\left(\frac{c_\kappa}{c_\kappa-1}\right)^{\xi^{(d)}}}{\alpha^{(d)}-2} + 1$.

B. Concentration: *there exist constants,* $\beta' > 1$ *and* $\xi' > 1$ *and* $1/2 \leq \eta' < 1$ *such that for* $n \geq 1$, $z \geq 1$,

$$\mathbb{P}(\tilde{v}^{(d-1)}(s_{i,d-1})_n - n\mu_*^{(d-1)}(s_{i,d-1}) \geq n^{\eta'}z) \leq \frac{\beta'}{z^{\xi'}},$$
$$\mathbb{P}(\tilde{v}^{(d-1)}(s_{i,d-1})_n - n\mu_*^{(d-1)}(s_{i,d-1}) \leq -n^{\eta'}z) \leq \frac{\beta'}{z^{\xi'}},$$

where $\eta' = \frac{\alpha^{(d)}}{\xi^{(d)}(1-\eta^{(d)})}$, $\xi' = \alpha^{(d)} - 1$, *and* β' *is a large enough constant that is function of parameters* $\alpha^{(d)}, \beta^{(d)}, \xi^{(d)}, \eta^{(d)}, \tilde{R}_{\max}^{(d-1)}, K, \Delta_{\min}^{(d-1)}(s_{i,d-1})$.

Completing Proof of Theorem 3.

Using Lemma 10, we conclude that the recursive relationship going from d to $d-1$ holds for all $d \geq 1$ with level 0 being the root. Thus, for the root state $s^{(0)}$, under n total simulations of annealed VOI-MCTS, the empirical average of the rewards $\frac{1}{n}\tilde{v}^{(0)}(s_0)_n$ is such that (using the fact that $\alpha^{(0)} \equiv \xi^{(0)}(1-\eta^{(0)})/\eta^{(0)}$)

$$\left|\mathbb{E}\left[\frac{1}{n}\tilde{v}^{(0)}(s_0)_n\right] - \mu_*^{(0)}(s_0)\right| \leq O\left(n^{\frac{\alpha^{(0)}}{\xi^{(0)}(1-\eta^{(0)})}-1}\right)$$
$$= O(n^{\eta-1}) \tag{60}$$

where $\mu_*^{(0)}(s_0)$ is the value for s_0 after D iterations of value iteration. We seek to transfer this root-level VOI-MCTS result to the VOIMCP estimator $\bar{V}_n(b_0) = \frac{1}{n}\sum_{a\in\mathcal{A}'} N(h_0a)\bar{Q}(h_0a)$.

The deterministic-transition analysis above extends to stochastic transitions by the same meta-node construction as in Lemma A.1 of Shah, Xie, and Xu (2022). Since $\mathcal{P}'$ is a finite-horizon POMDP, Lemmas 1 and 2 of Silver and Veness (2010) apply directly. Recall the derived history-MDP $\tilde{\mathcal{M}}'$ for $\mathcal{P}'$ defined in the proof of Theorem 2.

Let $\tilde{V}_D^*(h_0)$ denote the ordinary depth-D optimal value at the root of $\tilde{\mathcal{M}}'$. Since Lemma 10 is centered at the raw optimal mean, the root target satisfies

$$\mu_*^{(0)}(s_0) = \tilde{V}_D^*(h_0).$$

By Lemma 1 of Silver and Veness (2010), the value of any history-dependent policy in $\mathcal{P}'$ equals its value in the derived history-MDP $\tilde{\mathcal{M}}'$. Moreover, because the augmented action space $\mathcal{A}' = \mathcal{A}^{OL} \cup \mathcal{A}^{CL}$ contains a closed-loop counterpart of every original action and closed-loop backups weakly dominate open-loop backups, the ordinary optimal value in $\mathcal{P}'$ is exactly the original POMDP optimal value. Hence

$$\tilde{V}_D^*(h_0) = V_D^*(b_0).$$

By Lemma 2 of Silver and Veness (2010), the simulation traces generated by annealed VOIMCP in $\mathcal{P}'$ are equal in distribution to those generated by annealed VOI-MCTS in $\tilde{\mathcal{M}}'$. Therefore, the root estimator produced by annealed VOIMCP after n simulations has the same distribution as the root estimator analyzed above. Thus, we conclude that the output $\bar{V}_n(b_0)$ of annealed VOIMCP with n simulations satisfies

$$\left|\mathbb{E}[\bar{V}_n(b_0)] - V_D^*(b_0)\right| \leq \mathcal{O}(n^{\eta-1}), \tag{61}$$

thus proving the theorem.

Supplement on Empirical Evaluation

Implementation Details of VOIMCP

While the main text presents the theoretical recursive formulation of VOIMCP for convergence analysis, our practical implementation utilizes the standard PO-UCT expansion and rollout strategy for computational efficiency. Algorithm 2 details this implementation. The primary distinction is the termination of the SIMULATE procedure and returning a value estimate (e.g., with a heuristic rollout policy) in Line 17 upon expanding a new history node h.

Experimental Protocol

Each experimental trial refers to one full environment episode rollout under an online replanning framework, starting from an initial state/belief, and terminating under the task's terminal state or when the horizon is reached. We plan by interleaving policy planning and execution, using a standard SIR particle filter to update the belief before each replanning step. Each trial uses an independent seed, controlling any randomness in the trial, e.g., sampling from the generative model, particle filtering.

Hyperparameters

Table 1 summarizes the best-performing hyperparameters identified through grid search. We observed that VOIMCP is relatively robust to the exploration constant c. Across all three benchmark domains, we found that VOIMCP performance is relatively robust to the choice of κ, with values in the range $[0.01, 0.2]$ consistently outperforming the baselines.

Annealing Results

We additionally evaluate the performance of VOIMCP using a simplified count-based schedule where κ decays as $\kappa_N = \kappa_{max}\frac{x}{1+x}$ where $x = \frac{\kappa}{\kappa_{max}}B_N$. As visit counts grow, κ decays proportionally to B_N and is smoothly bounded in [0, κ_{max}). Figure 3 compares the discounted cumulative reward difference between the fixed-κ implementation and the annealed variant across all three benchmarks (annealed minus standard).

Algorithm 2: VOIMCP (Implementation)

procedure SEARCH(b)
  for all $i = 1, \ldots, n$ **do**
    $s \sim b$
    $h_0 = \{b\}$
    SIMULATE($s, h_0, 0$)
  end for
  return $\arg\max_{a \in \mathcal{A}'} \bar{Q}(ha)$
end procedure
procedure SIMULATE($s, h, depth$)
  if $depth = D$ **then**
    return 0
  end if
  if $h \notin T$ **then**
    for all $a \in \mathcal{A}'$ **do**
      $T(ha) \leftarrow (0, 0)$
    end for
    return VALUEESTIMATE($s, h, depth$)
  end if
  $a^* \in \arg\max_{a \in \mathcal{A}'} UCB_{VOI}(ha)$
  if $a^* \in \mathcal{A}_{CL}$ **then**
    $(s', o, r) \sim \mathcal{G}(s, a^*)$
  else
    $(s', r) \sim \mathcal{G}(s, a^*)$
    $o \leftarrow o_{null}$
  end if
  $N(h) \leftarrow N(h) + 1$; $N(ha^*) \leftarrow N(ha^*) + 1$
  $R \leftarrow r + \gamma$SIMULATE($s', ha^*o, depth{+}1$)
  $\bar{Q}(ha^*) \leftarrow \bar{Q}(ha^*) + \dfrac{R - \bar{Q}(ha^*)}{N(ha^*)}$
  return R
end procedure

Domain	Algorithm	c	κ	D
Tracking	VOIMCP	100	0.03	20
	PO-UCT	100	-	20
	I-UCB POMCP	100	-	20
	Open-Loop	100	-	20
FVRockSample	VOIMCP	10	0.02	20
	PO-UCT	10	-	20
	I-UCB POMCP	10	-	20
	Open-Loop	10	-	20
Laser Tag	VOIMCP	100	0.01	20
	PO-UCT	100	-	20
	I-UCB POMCP	100	-	20
	Open-Loop	100	-	20

Table 1: Summary of hyperparameters used in experiments.

These results show that this annealing schedule is not only theoretically rigorous but also practically beneficial. Over a range of κ_{max}, when compared to a fixed κ_{max} baseline, the annealed VOIMCP variant achieved peak performance improvements of approximately 2.52%, 4.06%, and 3.44%

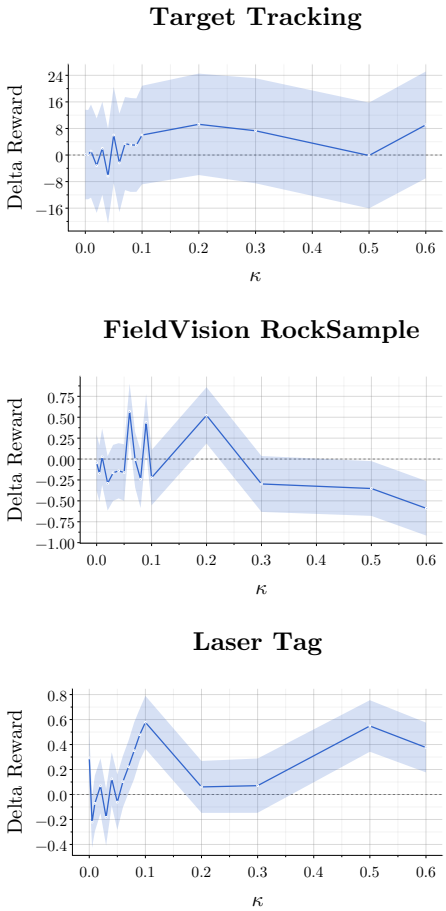

Figure 3: Results for VOIMCP difference between the annealed and standard algorithms, presenting delta discounted cumulative reward over 1000 trials. The lighter-colored ribbons around the Monte Carlo mean display the 95% confidence interval.

for the Target Tracking, FieldVision RockSample, and Laser Tag domains, respectively. While these gains are modest, these results serve as a valuable proof of concept: annealing can provide a path to asymptotic optimality while reducing reliance on careful tuning of a fixed κ. A more exhaustive analysis of complex annealing schedules and their empirical trade-offs remains a subject for future work.